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202220953 신현준

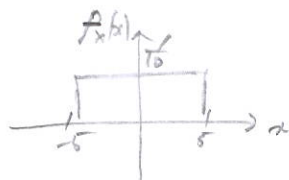
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Problem 1. (10 points)

Let X be a continuous random variable uniformly distributed over $(-5, 5)$, i.e., $X \sim \text{Unif}(-5, 5)$. Answer the following questions.

- What is the probability density function (PDF) $f_X(x)$?
- What is the cumulative distribution function (CDF) $F_X(x)$?
- What is $\mathbb{E}[X]$?
- What is $\mathbb{E}[X^5]$?
- What is $\mathbb{E}[e^X]$?

(a)



$$f_X(x) = \begin{cases} \frac{1}{10} & (-5 \leq x \leq 5) \\ 0 & (\text{otherwise}) \end{cases}$$

(b)

$$F_X(x) = \int_{-\infty}^x f_X(\tau) d\tau = \begin{cases} 0 & (x < -5) \\ \frac{1}{10}(x+5) & (-5 \leq x \leq 5) \\ 1 & (x > 5) \end{cases}$$

$$(c) \mathbb{E}[X] = \int_{-5}^5 \underbrace{x}_{\text{odd}} \underbrace{\frac{1}{10}}_{\text{even}} dx = 0$$

$$(d) \mathbb{E}[X^5] = \int_{-5}^5 \underbrace{x^5}_{\text{odd}} \underbrace{\frac{1}{10}}_{\text{even}} dx = 0$$

$$(e) \mathbb{E}[e^X] = \int_{-5}^5 e^x \cdot \frac{1}{10} dx = \left[\frac{1}{10} e^x \right]_{-5}^5 = \frac{1}{10} (e^5 - e^{-5})$$

Problem 2. (20 points)Random variables X and Y have the following joint PDF:

$$f_{X,Y}(x,y) = \begin{cases} \frac{5x^2}{2}, & -1 \leq x \leq 1, 0 \leq y \leq x^2, \\ 0, & \text{otherwise.} \end{cases}$$

Answer the following questions.

(a) What are $\mathbb{E}[X]$ and $\text{Var}[X]$?(b) What are $\mathbb{E}[Y]$ and $\text{Var}[Y]$?(c) What is $\text{Cov}[X, Y]$?(d) What is $\mathbb{E}[X + Y]$?(e) What is $\text{Var}[X + Y]$?

(a) $f_X(x) = \int_0^{x^2} \frac{5x^2}{2} dy = \frac{5x^2}{2} \cdot x^2 = \frac{5x^4}{2}, (-1 \leq x \leq 1, \text{ otherwise } 0)$

$$\mathbb{E}[X] = \int_{-1}^1 x \cdot \frac{5x^4}{2} dx = 0 \quad \mathbb{E}[X^2] = \int_{-1}^1 x^2 \cdot \frac{5x^4}{2} dx = \int_0^1 5x^6 dx = \frac{5}{7}$$

$$\text{Var}[X] = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = \frac{5}{7}$$

(b) $f_Y(y) = \int_{-1}^1 \frac{5x^2}{2} dx = \int_0^{\sqrt{y}} 5x^2 dx = \frac{5}{3} \quad (0 \leq y \leq 1, \text{ otherwise } 0)$

$$\mathbb{E}[Y] = \int_0^1 y \cdot \frac{5}{3} dy = \frac{5}{6} \quad \mathbb{E}[Y^2] = \int_0^1 y^2 \cdot \frac{5}{3} dy = \frac{5}{9}$$

$$\text{Var}[Y] = \mathbb{E}[Y^2] - (\mathbb{E}[Y])^2 = \frac{5}{9} - \frac{25}{36} = \frac{20}{36} = \frac{5}{9}$$

(c) $\mathbb{E}[XY] = \int_{-1}^1 \int_0^{x^2} xy \cdot \frac{5x^2}{2} dy dx = \int_{-1}^1 \frac{5x^3}{2} \cdot \frac{1}{2} x^4 dx = 0$

$$\text{Cov}(X,Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y] = 0 - 0 \cdot \frac{5}{6} = 0$$

(d) $\mathbb{E}[X+Y] = \mathbb{E}[X] + \mathbb{E}[Y] = 0 + \frac{5}{6} = \frac{5}{6}$

(e) $\text{Var}[X+Y] = \text{Var}[X] + \text{Var}[Y] + 2\text{Cov}(X,Y)$

$$= \frac{5}{7} + \frac{5}{9} + 2 \cdot 0 = \frac{50}{63}$$

Problem 3. (10 points)

The moment generating function (MGF) of a random variable Y is defined as

$$M_Y(s) \triangleq \mathbb{E}[e^{sY}].$$

Obtain the MGF of the following Gaussian random variable with parameters $\mu = 0$ and $\sigma^2 = 1$:

$$f_Y(y) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{y^2}{2}\right), \quad -\infty < y < \infty.$$

$$\int_{-\infty}^{\infty} \exp\left(-\frac{y^2}{2}\right) dy = \sqrt{2\pi}$$

$$\begin{aligned} E[e^{sY}] &= \int_{-\infty}^{\infty} e^{sy} \cdot f_Y(y) dy \\ &= \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} \exp(sy) \cdot \exp\left(-\frac{y^2}{2}\right) dy \\ &= \frac{1}{\sqrt{2\pi}} e^{s^2/2} \cdot \Gamma(1) \Big|_{-\infty}^{\infty} \end{aligned}$$

Problem 4. (10 points)

The correlation coefficient between two random variables X and Y is defined as

$$\rho_{X,Y} = \frac{\text{Cov}(X,Y)}{\sqrt{\text{Var}(X)}\sqrt{\text{Var}(Y)}}$$

(a) Let X and Y be discrete random variables with the following joint probability mass function (PMF):

	$Y = -1$	$Y = 0$	$Y = 1$	
$X = -1$	$\frac{1}{8}$	$\frac{1}{8}$	0	$2 \rightarrow P(X=-1) = \frac{2}{8}$
$X = 0$	$\frac{1}{8}$	$\frac{1}{4}$	$\frac{1}{8}$	$4 \rightarrow P(X=0) = \frac{4}{8}$
$X = 1$	0	$\frac{1}{8}$	$\frac{1}{8}$	$2 \rightarrow P(X=1) = \frac{2}{8}$

Find the correlation coefficient $\rho_{X,Y}$.

(b) Let X be a continuous random variable uniformly distributed over $(-1, 1)$, i.e., $X \sim \text{Unif}(-1, 1)$. Define another random variable Y by $Y = X^2$. Find the correlation coefficient $\rho_{X,Y}$.

(a) $E[X] = (-1) \cdot \frac{2}{8} + 1 \cdot \frac{2}{8} = 0$ $E[X^2] = \frac{1}{8} + \frac{2}{8} = \frac{1}{2}$ $\text{Var}[X] = \frac{1}{2}$
 $E[Y] = (-1) \cdot \frac{2}{8} + 1 \cdot \frac{2}{8} = 0$ $E[Y^2] = \frac{2}{8} + \frac{2}{8} = \frac{1}{2}$ $\text{Var}[Y] = \frac{1}{2}$
 $E[XY] = (-1)(-1) \frac{1}{8} + (1)(1) \frac{1}{8} = \frac{2}{8} = \frac{1}{4}$

$\text{Cov}(X,Y) = E[XY] - E[X]E[Y] = \frac{1}{4} - 0 \cdot 0 = \frac{1}{4}$

$\rho_{X,Y} = \frac{\text{Cov}(X,Y)}{\sqrt{\text{Var}(X)}\sqrt{\text{Var}(Y)}} = \frac{\frac{1}{4}}{\sqrt{\frac{1}{2} \cdot \frac{1}{2}}} = \frac{\frac{1}{4}}{\frac{1}{2}} = \frac{1}{2}$

(b) $f_X(x) = \begin{cases} \frac{1}{2} & (-1 \leq x \leq 1) \\ 0 & (\text{otherwise}) \end{cases}$ $E[X] = 0$ $E[Y] = E[X^2] = \int_{-1}^1 x^2 \cdot \frac{1}{2} dx = \int_0^1 x^2 dx = \frac{1}{3}$

$E[XY] = E[X^3] = \int_{-1}^1 x^3 \cdot \frac{1}{2} dx = 0$

$\text{Cov}(X,Y) = E[XY] - E[X]E[Y] = E[X^3] - E[X]E[Y] = 0 - 0 \cdot \frac{1}{3} = 0$

$\rho = \frac{\text{Cov}(X,Y)}{\sqrt{\text{Var}(X)}\sqrt{\text{Var}(Y)}} = 0$

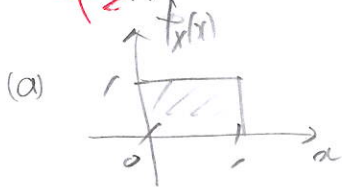
Problem 5. (10 points)

Let X and Y be independent random variables, each uniformly distributed over $(0, 1)$, i.e., $X \sim \text{Unif}(0, 1)$ and $Y \sim \text{Unif}(0, 1)$. Answer the following questions.

(a) Define a new random variable Z by $Z = X^2$. Find the PDF $f_Z(z)$ of Z .

(b) Define a new random variable S by $S = X + Y$. Find the PDF $f_S(s)$ of S .

(c) Define a new random variable W by $W = XY$. Find the PDF $f_W(w)$ of W .



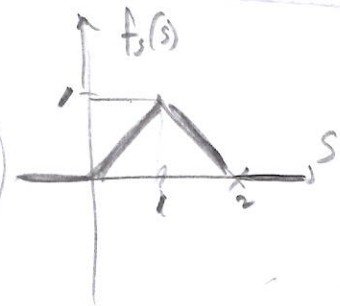
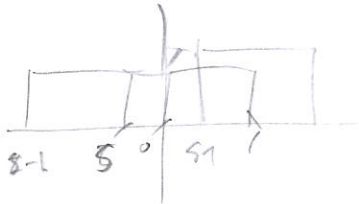
$$F_Z(z) = P(Z \leq z) = P(X^2 \leq z) = P(0 \leq X \leq \sqrt{z}) \begin{cases} 0 & (0 \leq z) \\ \sqrt{z} & (0 \leq z \leq 1) \\ 1 & (z > 1) \end{cases}$$

$$= \int_0^{\sqrt{z}} 1 \, dx = \sqrt{z} \quad (0 \leq z \leq 1) \rightarrow$$

$$f_Z(z) = \frac{dF_Z(z)}{dz} = \frac{1}{2} z^{-\frac{1}{2}} = \frac{1}{2\sqrt{z}} \quad (0 \leq z \leq 1, \text{ otherwise } 0).$$

(b) X 과 Y 가 독립이므로 $f_S(s) = f_X(x) * f_Y(y)$

$$f_S(s) = \int_{-\infty}^{\infty} f_X(x) f_Y(s-x) \, dx = \begin{cases} 0 & (s < 0) \\ s & (0 \leq s < 1) \\ 2-s & (1 \leq s < 2) \\ 0 & (s \geq 2) \end{cases}$$



(c) $f_W(w) = P(W \leq w) = P(XY \leq w) = \begin{cases} 0 & (w < 0, w > 1) \end{cases} \leftarrow \text{일단 이거는 알겠다.}$

$$P\left(X \leq \frac{w}{y}\right) P(Y=y)$$

$\frac{w}{y} \quad f_Y(y)$

Problem 6. (20 points)

In a wireless communication receiver, the instantaneous channel power gain H under Rayleigh fading is often modeled as an exponential random variable. Suppose $H \sim \text{Exp}(\lambda)$, where $\lambda > 0$, with PDF

$$f_H(h) = \begin{cases} \lambda e^{-\lambda h}, & h \geq 0, \\ 0, & \text{otherwise.} \end{cases}$$

The received energy measurement is modeled as $Y = h_0 H + N$, where $h_0 > 0$ is a deterministic transmit-power-related constant, and N is an independent noise energy term satisfying $N \sim \text{Exp}(\mu)$, where $\mu > 0$. The PDF of N is

$$f_N(n) = \begin{cases} \mu e^{-\mu n}, & n \geq 0, \\ 0, & \text{otherwise.} \end{cases}$$

$$\rightarrow F_N(n) = 1 - e^{-\mu n}$$

Answer the following questions.

(a) Given $H = h$, find the conditional PDF $f_{Y|H}(y|h)$. **3**

(b) Compute $\mathbb{E}[Y|H]$ and $\text{Var}[Y|H]$. **5**

(c) Compute the unconditional mean $\mathbb{E}[Y]$.

(d) Compute the unconditional variance $\text{Var}[Y]$.

(a) $H=h$ 일 때 $P(Y \leq y | H=h) = P(N + h h_0 \leq y) = P(N \leq y - h h_0)$
 $= 1 - e^{-\mu(y - h h_0)}$ **3**

(b) $\mathbb{E}[Y|H] = \mathbb{E}[h_0 H + N] = h_0 H + \mathbb{E}[N] = h_0 H + \frac{1}{\mu}$ **5**
 $\text{Var}[Y|H] = \text{Var}[h_0 H + N] = \text{Var}[N] = \frac{1}{\mu^2}$ **5**

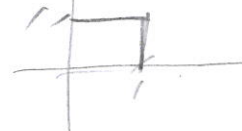
(c) $\mathbb{E}[Y] = \mathbb{E}[\mathbb{E}[Y|H]] = \mathbb{E}[h_0 H + \frac{1}{\mu}] = h_0 \mathbb{E}[H] + \mathbb{E}[\frac{1}{\mu}]$ **3**

(d) $\text{Var}[Y] = \mathbb{E}[\text{Var}(Y|H)] + \text{Var}[\mathbb{E}[Y|H]]$
 $= \mathbb{E}[\frac{1}{\mu^2}] + \text{Var}[h_0 H + \frac{1}{\mu}]$
 $= \mathbb{E}[\frac{1}{\mu^2}] + \text{Var}[\frac{1}{\mu}]$ **5**

Problem 7. (20 points)

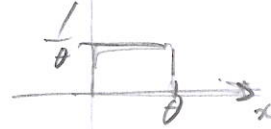
Let Θ be an unknown random parameter representing the upper limit of a uniform distribution. Suppose that the prior distribution of Θ is $\Theta \sim \text{Unif}(0, 1)$, that is,

$$f_{\Theta}(\theta) = \begin{cases} 1, & 0 < \theta < 1, \\ 0, & \text{otherwise.} \end{cases}$$



Given $\Theta = \theta$, the observation X is uniformly distributed over $(0, \theta)$, i.e., $X|\Theta = \theta \sim \text{Unif}(0, \theta)$. Therefore,

$$f_{X|\Theta}(x|\theta) = \begin{cases} \frac{1}{\theta}, & 0 < x < \theta, \\ 0, & \text{otherwise.} \end{cases}$$



Assume that a single observation $X = x$ is given, where $0 < x < 1$. Answer the following questions.

- Using Bayes' rule, find the posterior PDF $f_{\Theta|X}(\theta|x)$.
- Find the maximum a posteriori (MAP) estimator of Θ , defined by $\hat{\theta}_{\text{MAP}}(x) = \arg \max_{\theta} f_{\Theta|X}(\theta|x)$. If the posterior density is maximized at a boundary, use the corresponding boundary value as the MAP estimate.
- Find the least mean square (LMS) estimator of Θ , defined by $\hat{\theta}_{\text{LMS}}(x) = \mathbb{E}[\Theta|X = x]$.
- Consider a linear LMS estimator of the form $\hat{\Theta}_L = aX + b$. Find the constants a and b that minimize $\mathbb{E}[(\Theta - aX - b)^2]$.

(a) $f_{\Theta|X} = \frac{f_{\Theta}(\theta) \cdot f_{X|\Theta}(x|\theta)}{f_X(x)} = \frac{1}{\theta} \cdot \frac{1}{f_X(x)} \quad \left(f_X(x) \text{ is } \frac{1}{x} \text{ for } x < \theta < 1 \right)$

$f_X(x) = \int_x^1 \frac{1}{\theta} d\theta = \ln(1) - \ln(x) = -\ln(x) = \ln\left(\frac{1}{x}\right)$

$\rightarrow f_{\Theta|X} = \frac{1}{\theta \ln\left(\frac{1}{x}\right)}$ 정답!

(c) $\theta = \mathbb{E}[\Theta | X = x]$

(d) $a = \frac{\text{cov}(X, \Theta)}{\text{var}(X)}$ $b = \mathbb{E}[\Theta] - a\mathbb{E}[X]$

Additional answer space for Problem 7